Do Now · retrieval

Thinking abstractly: models and reality

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is unit testing?

Q2

What is a precondition?

Q3

What do vertices and edges represent in a graph?

Q4

Why is an unweighted route graph not enough to identify the fastest journey?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Choose a purpose — Decide whether a particular book may be loaned.

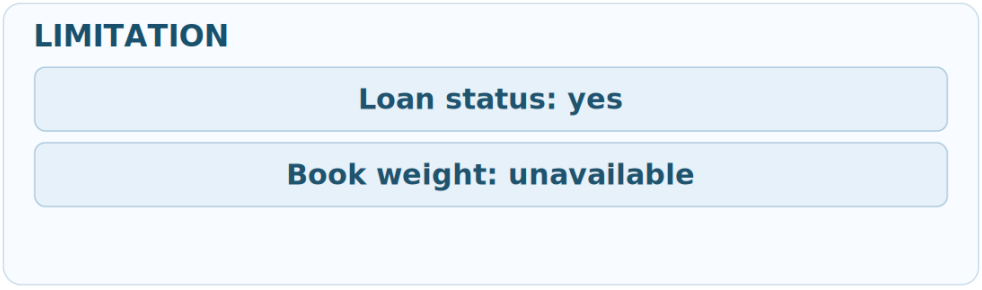
Core idea 2

Keep essential facts — ISBN identifies the book, and Available indicates whether it can be borrowed.

Core idea 3

Evaluate omissions — The model cannot answer questions about book weight or appearance because those attributes are omitted.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

abstraction — Retain relevant characteristics, ignore irrelevant ones.

model — Simplified representation serving a purpose.

attribute — Property stored about an entity.

assumption — Condition accepted in constructing the model.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Model a bus network for finding the route with the fewest stops.

Follow the worked method

1. Predict first — Model each bus stop as a vertex and each permitted connection as an edge.

2. Model and check — Vehicle colour is irrelevant. To find fastest rather than fewest-stop routes, add journey-time weights.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Only valid marks and the number of entries are needed to compute the mean.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Identify two relevant and two irrelevant attributes for a stock checker.

Question 2

Give a simple abstraction for detecting school room timetable clashes: three needed fields and two omitted details.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Why should a model not contain every real-world detail? Give an example where a previously omitted detail becomes relevant.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A good abstraction must record every real-world property.

Use key vocabulary from memory

1. Explain abstraction in your own words.

2. Explain model in your own words.

3. Explain attribute in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.